

Instructor Key: Day 5 Mastery Quiz

Boolean Combinations

Key for the Day 5 student quiz package.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 5 mastery quiz; quiz, not workbooklet.
Content anchor	Built after Lance approved the Day 5 workbooklet content surface: Boolean combinations, Week 1 synthesis, support diagnosis, and no branch routing yet.
Difficulty posture	Highest daily quiz in the Day 3–5 sequence. It combines Day 2 current-state habit, Day 3 cast/truthiness discipline, Day 4 boundary evidence, and Day 5 and/or/not reasoning.
Scenario/source boundary	Lower-sensitivity pump readiness record. This remains within Week 1; Lance’s new rule is that the Week 1 larger assessment should close the pump scenario and Week 2 should start a new scenario family.
Student-surface rule	Student PDF intentionally hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.
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2 Reference code

```
1 # converted values and setup evidence
2 flow_text = "21.6"
3 flow_lpm = float(flow_text)
4 low_limit_lpm = 20.4
5 high_limit_lpm = 21.6
6 approval_label = "reviewed"
7 operator_initials = ""
8 retry_count = 0
9 sensor_ready = True
10
11 # separate evidence checks
12 lower_ok = flow_lpm >= low_limit_lpm
13 upper_ok = flow_lpm <= high_limit_lpm
14 flow_ok = lower_ok and upper_ok
15 approval_ok = approval_label == "reviewed"
16 setup_ok = sensor_ready and operator_initials != ""
17 no_retry_needed = retry_count == 0
18 record_ready = flow_ok and approval_ok and setup_ok and no_retry_needed
19 needs_followup = (not flow_ok) or (not approval_ok) or (not setup_ok)
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Common issue / support move	Tags
1	Separate boundary evidence	<code>lower_ok = True; upper_ok = True</code> . The upper check is true because <code>21.6 <= 21.6</code> includes equality.	If upper is false, revisit Day 4 inclusive upper boundary.	trace; predict
1	and for flow	<code>flow_ok = True</code> ; created by <code>lower_ok</code> and <code>upper_ok</code> . Both parts are true.	If student says or , point to the line assigning <code>flow_ok</code> .	trace; explain
1	Explicit label check	<code>approval_ok = True</code> . It uses <code>approval_label == "reviewed"</code> , an exact comparison, not <code>bool(approval_label)</code> .	If student says truthiness, return to Day 3 label-check distinction.	trace; explain
1	Setup check fails	<code>setup_ok = False</code> . <code>sensor_ready</code> is true, but <code>operator_initials != ""</code> is false because initials are blank.	Key synthesis trap: one false part makes and false.	debug; explain
1	Retry check	<code>no_retry_needed = True; retry_count == 0</code> .	If student uses truthiness, ask for the exact equality check.	predict
1	Final readiness	<code>record_ready = False</code> . Required condition preventing readiness: <code>setup_ok</code> .	Require naming the controlling failed condition, not only final false.	debug; explain
1	Follow-up route	<code>needs_followup = True</code> . The controlling true term is not <code>setup_ok</code> .	If student says flow or approval, have them compute each <code>not(...)</code> term.	debug; explain
1	Recompute after initials	If <code>operator_initials = "AM"</code> and all evidence is recomputed, <code>setup_ok = True</code> and <code>record_ready = True</code> .	Make sure students recompute, not merely change the final variable by hand.	transfer; trace
1	Unsafe or claim	The note uses or when the rule requires all readiness evidence. Flow and approval are not enough because setup evidence fails: blank operator initials.	Require connector critique plus missing setup condition.	debug; test
1	Correct combined expression	<code>flow_ok</code> and <code>approval_ok</code> and <code>setup_ok</code> and <code>no_retry_needed</code> . Equivalent parenthesized all-required forms are acceptable.	If student uses or , ask whether any one passing evidence is sufficient for readiness.	implement; transfer

4 Expected Boolean trace

Name/expression	Value	Reason
<code>lower_ok</code>	True	<code>21.6 >= 20.4</code> .
<code>upper_ok</code>	True	<code>21.6 <= 21.6</code> ; exact upper limit included.
<code>flow_ok</code>	True	Both lower and upper checks are true.
<code>approval_ok</code>	True	Exact label comparison succeeds.
<code>setup_ok</code>	False	Sensor is ready, but operator initials are blank.
<code>no_retry_needed</code>	True	Retry count equals zero.
<code>record_ready</code>	False	All-required and ; <code>setup_ok</code> is false.
<code>needs_followup</code>	True	not <code>setup_ok</code> is true.

5 Mastery interpretation and support routing

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10, including controlling false condition and follow-up term	Student can trace separate evidence, combine with and , use not for follow-up, and diagnose unsafe or .	Ready for Week 1 larger evidence event closing the pump scenario.
Developing	5–7 / 10, or final result correct without cause	Student can compute some Boolean values but may not identify which condition controls the result.	Use a condition ledger: source, value, connector, final effect.
Needs support	0–4 / 10	Student does not yet preserve separate evidence before combining conditions.	Return to Day 4 separate lower/upper checks, then combine only two booleans at a time.

6 Optional recovery mini-surface

Use if several students treat **or** as if it meant all conditions passed.

```
a = True
b = False
a and b # False: one required part failed
a or b  # True: at least one part passed
not b   # True: b failed, so follow-up may be needed
```

Secure recovery answer: **and** is the all-required connector; **or** is the at-least-one connector; **not** reverses a Boolean value for follow-up logic.

7 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, print-safe margins, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores. Working release threshold remains Ally at least 95% and Panorama at least 95%.